

Cybersecurity Fundamentals and Defensive Engineering

The Core Security Paradigm: The CIA Triad

Defensive information security is anchored by the CIA triad:

1. Confidentiality: Ensuring that sensitive information is accessible solely to authorized personnel and protected against unauthorized disclosure.
2. Integrity: Safeguarding the accuracy, completeness, and tamper-resistance of data across its lifecycle, in transit and at rest.
3. Availability: Guaranteeing timely and dependable access to computing infrastructure and critical business data for authorized users.

Threat Vectors and the Modern Attack Surface

Contemporary cyber threat actors deploy multifaceted intrusion tactics:

- Ransomware: Advanced persistent threat (APT) groups infiltrate corporate perimeters, exfiltrate confidential databases, and systematically encrypt storage volumes to demand extortion payments.
- Social Engineering and Spear Phishing: Highly targeted spear-phishing campaigns leverage OSINT and synthetic media to bypass traditional email filters and harvest privileged credentials.
- Zero-Day Vulnerabilities: Unknown software vulnerabilities exploited prior to developer patch availability. Rigorous open-source intelligence (OSINT) and bug bounty incentives help identify these gaps before malicious weaponization.

Cryptographic Foundations

Cryptography underpins digital trust. Symmetric encryption algorithms like AES-256 deliver high-throughput protection for bulk data storage. Asymmetric algorithms (such as RSA and Elliptic Curve Cryptography) enable public key infrastructure (PKI), digital signatures, and secure key exchanges.

Transport Layer Security (TLS 1.3) protects communication over untrusted networks using ephemeral Diffie-Hellman key exchanges to maintain forward secrecy.

Zero Trust Architecture (ZTA)

The historic castle-and-moat security model assumed perimeter defense was sufficient. In contrast, Zero Trust operates under the fundamental principle of 'never trust, always verify.' Every access request?regardless of origin within the internal intranet or outside?undergoes continuous mutual authentication, fine-grained role-based access control (RBAC), and behavioral anomaly evaluation before permissions are granted.